

Discussion

The current study aimed to examine the relationship between intellectual humility and apology quality, and whether a post-transgressional response also impacted apology quality. In accordance with our preliminary analysis, we did find a positive correlation between intellectual humility and apology quality. However, our primary analyses did not support our hypotheses. We found there was no significant effect of intellectual humility on apology quality. Furthermore, apology quality was unaffected by a victim's post-transgressional response. Finally, the interaction between intellectual humility and post-transgressional responses also did not affect apology quality. These findings do not undermine the strength of our study, which utilizes experimental design. This goes beyond the limitations of only using a correlational design.

Our study found a positive correlation between intellectual humility and apology quality. Ludwig et al., (2022) also found there to be a statistically significant relationship between intellectual humility and apology comprehensiveness. This can be explained as individuals who are higher in intellectual humility are more likely to be aware of their own limits and admit when they are wrong, leading to a higher chance of apologizing in a genuine way. Combining these findings, because both experiments found a positive association, there is a strengthened likelihood that the two variables are in fact related. This consistency increases our confidence in the relationship between the variables. Additionally, these results confirm our belief that the vignettes used in both studies were temporarily influencing intellectual humility as intended.

Next, when looking at the main effect of intellectual humility on apology quality. Our study aimed to heighten one's intellectual humility by having them imagine a scenario where they acted in accordance with the definition of intellectual humility. This manipulation of intellectual humility did not have a statistically significant effect on apology quality. The present

results are consistent with Koetke et al.'s (2023) work, which deals with the effect of intellectual humility on investigative behavior. Their manipulation of intellectual humility, using the fallibility salience test, successfully increased participants' levels of intellectual humility, similar to our study. However, their study did not find a direct effect between intellectual humility and investigative behavior. This was explained because of the minimal manipulation of the fallibility salience test. We conclude that our method of manipulation suffered a similar challenge, where although we were able to successfully manipulate intellectual humility, it was not enough to produce a direct effect with apology quality. Perhaps, using a stronger manipulation test in our study would have led to statistically significant results for this main effect. An example of this could potentially be conducting the study under experimental conditions where the manipulation was done with other real participants or confederates.

Additionally, we found there to be no statistically significant relationship between post-transgression response and apologies. Struthers and colleagues (2008) also looked at the effect of a victim's post-transgression response on a transgressor's repentance using a laboratory experiment. Participants in this experiment acted as a tutor who provided answers to a student. The study was rigged by having the participants give answers that were incorrect to the student. This induced a transgression, causing the student to not complete his assignment. After this manipulation, the student responded in one of four ways and the participant's repentance was measured. Repentance is the act of turning away from a wrong and changing oneself to not repeat that wrong. In comparison to an apology, it goes beyond just addressing the wrongdoing. Struthers and colleagues (2008) instructed participants to provide their partners (the student) with feedback based on their interactions. This feedback was coded by independent raters. Participants were also tasked with grading their partner on a seven-point scale. Finally, they

measured repentance using a different seven-point scale. Their results revealed a significant effect of forgiveness on repentance. This is a stark contrast compared to our study. However, there are a couple explanations that can explain these differences. Struthers and colleagues (2008) had students respond in four different ways, while we only had two forgiveness conditions. Furthermore, their method included two additional surveys where the participants addressed the victim's post-transgression response, whereas we only had participants rank how they themselves would respond back to the victim. Perhaps an even greater explanation, Struthers and colleagues (2008) had participants interact with a real person. Our study had participants imagine themselves in a scenario that they read, with no actual interactions. Overall, these differences paint a clear picture of why we may not have found a significant main effect of post-transgression response on apology quality.

Finally, we delved into the interaction between intellectual humility and post transgression response on apology quality. Again, we found an insignificant interaction between these two variables. This means regardless of the post-transgression response of the victim, when placed in the intellectual humility condition, there was no significant effect on their apology quality. A possible explanation for this finding is that participants may have over- or under-reacted compared to how they would have reacted in the specific scenario we had them imagine themselves in. This helps us understand why we did not find any significant relationships between intellectual humility and post-transgression response on apology quality.

Two primary limitations were identified in our method. First, in terms of sampling, a convenience sampling method was used, which does not provide a representative sample of the general population. Thus, the results we found may not be applicable to other groups and so we

are unable to draw a broader conclusion based on these results. We could address this limitation by using a representative sample in the future.

Secondly, our measure of the dependent variable was heavily reliant on self-report measures. These measures are always subject to external uncontrolled variables like environmental factors, as well as internal factors such as social desirability. As a result, our procedure lacked internal validity. This limitation could be addressed by conducting a future study in a lab environment, where increased internal validity would allow for better control of these variables. Creating a simulated environment where participants cannot self-report but instead have their behavior constantly monitored could produce more accurate results.

In future research, we would like to investigate the relationship between intellectual humility and admission of guilt. Although our research did not show a direct relationship between intellectual humility and apology quality, previous research has demonstrated there to be a link between intellectual humility and wrongdoing admission. It would be interesting to identify if this relationship transitions into a courtroom simulation.

Additionally, looking at how demographic differences between the transgressor and the victim could produce interesting results. For example, does post-transgressional response and/or apology quality change when the victim and transgressor are of different sex, age, or race? We hypothesize that social psychology factors, such as social identity theory, may result in decreased intellectual humility and/or apology quality to members of different groups than within groups.

References

Ludwig, J. M., Schumann, K., & Porter, T. (2022). Humble and apologetic? Predicting apology quality with intellectual and general humility. *Personality and Individual Differences*, 188, 111477. <https://doi.org/10.1016/j.paid.2021.111477>

Koetke, J., Schumann, K., Porter, T., & Smilo-Morgan, I. (2023). Fallibility Salience Increases Intellectual Humility: Implications for People's Willingness to Investigate Political Misinformation. *Personality & social psychology bulletin*, 49(5), 806–820. <https://doi.org/10.1177/01461672221080979>

Koetke, J., Schumann, K., Welker, K., & Coleman, P. T. (2024). Intellectual humility is reliably associated with constructive responses to conflict. *PLoS ONE*, 19(9), 12. <https://doi.org/10.1371/journal.pone.0309848>

Struthers, C. W., Eaton, J., Shirvani, N., Georghiou, M., & Edell, E. (2008). The Effect of Preemptive Forgiveness and a Transgressor's Responsibility on Shame, Motivation to Reconcile, and Repentance. *Basic and Applied Social Psychology*, 30(2), 130–141. <https://doi.org/10.1080/01973530802209178>